MATH INQUIRERS HOLIDAY PROGRAM

Free 30-minute taster

Lesson 1: Factor Hunters (Factors and multiples) • Math Inquirers Level 4 (ages 9-11)

Notice it. Test it. Explain why.

What this is

A short sample of a real Math Inquirers lesson for Level 4 (ages 9-11, covers P4 topics). It is Lesson 1 of the Level 4 holiday program (Factor Hunters), shortened to about 30 minutes: a warm-up, a hands-on explore, a mini-lesson, a few worksheet questions and an exit ticket. The same routine runs through all ten 90-minute lessons.

The 30 minutes

Time	Part	What happens
5 min	**Warm-up**	Fizz-Buzz: count round the group from 1. Say "Fizz" for a multiple of 3, "Buzz" for a multiple of 4 and "Fizz-Buzz" for a multiple of both. Which numbers were Fizz-Buzz?
8 min	**Explore**	Pairs use 24 square tiles to build every different rectangle that uses all the tiles (a 3 by 8 and an 8 by 3 count as the same). They record each one as a multiplication and decide how they know they have found them all.
5 min	**Mini-lesson**	Factor pairs in order: 36 = 1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6 (5 does not divide 36); stop when the pairs meet and draw a factor rainbow. Common factors: 12 apples and 18 oranges share the factors 1, 2, 3, 6, so 6 bags with 2 apples and 3 oranges each. Common multiples: buses every 6 and 8 minutes meet again after 24 minutes. Before page 2: square numbers have an odd number of factors (100 has 9; 18 has 6); for riddles, list the multiples of the bigger number first, then check each clue.
9 min	**Worksheet**	The child works on questions 1 to 4 of the Lesson 1 sheet (next pages), then checks them with the tutor.
3 min	**Exit ticket**	Write all the factors of 28. Then find the smallest number that is a multiple of both 6 and 9.

What you need

36 square tiles or squared paper per pair, mini whiteboards, coloured pencils, a 1 to 50 number strip.

Online: Children cut 24 paper squares (or use squared paper) before the lesson and build rectangles on camera; the tutor marks the bus time line and the factor rainbow on the slides.

Pages in this file: this page, the child's worksheet (about two pages), then tutor notes with the answers. Do not give the tutor-notes pages to the child. This taster is enrichment for thinking skills; it does not promise exam results and is not affiliated with MOE or the Gifted Education Programme.

FACTOR HUNTERS

Name _____ Group _____ Date _____

No.1

Hunt down every factor of 48. (A factor goes into 48 exactly.)

a Write the factor pairs of 48 in order. You may not need every box.

1 × 48	__ × __	__ × __	__ × __	__ × __	__ × __

b List all the factors of 48 in order.

Factors: _____

c How many factors does 48 have? _____ How did you know when to stop looking?

.....

No.2

A gardener plants 40 trees in equal rows. There are at least 2 rows, and at least 2 trees in each row. (2 rows of 20 trees and 20 rows of 2 trees are different ways to plant.)

a Find every way. One way is 4 rows of 10 trees. Fill in the table. You may not need every column.

Rows	4						
Trees in each row	10						

How many ways? _____

No.3

Mei has 18 red beads and 24 blue beads. She puts them into identical packs. Every pack has the same number of red beads and the same number of blue beads. No beads are left over.

a Factors of 18: _____ Factors of 24: _____

b Common factors of 18 and 24: _____

c Greatest number of packs: _____ Each pack has _____ red and _____ blue.

No.4

At an interchange, Bus 33 leaves every 10 minutes and Bus 51 leaves every 15 minutes. Both buses leave at 6:00 am.



a Bus 33: 6:00, 6:10, _____, _____ ... Bus 51: 6:00, _____, _____ ...

They next leave together at: _____

b How many times do they leave together from 6:00 am to 8:00 am? Count 6:00 and 8:00 too.

Number of times: _____

Tutor notes: Lesson 1, Factor Hunters

Factors and multiples • Level 4 (ages 9-11, covers P4 topics)

Goal

Find all the factor pairs of a number by working in order, find common factors and common multiples to solve sharing and timetable puzzles, discover which numbers have an odd number of factors, and solve multiple-based logic riddles.

Skills: Making a list in order, looking for a pattern, using clues one at a time, explaining why

Opening

Welcome, Factor Hunters! Every number hides its factors inside it, and today we hunt them all down, without missing a single one.

We will also find out when two buses meet again, and why some numbers have an odd number of factors. Notice it. Test it. Explain why.

Warm-up (about 5 minutes)

Fizz-Buzz: count round the group from 1. Say "Fizz" for a multiple of 3, "Buzz" for a multiple of 4 and "Fizz-Buzz" for a multiple of both. Which numbers were Fizz-Buzz?

We count round the group, one number each. If your number is in the 3 times table, say "Fizz". In the 4 times table, say "Buzz". In both, say "Fizz-Buzz"!

If you slip, no problem: say "My go again" and pick someone to start the next round.

Before we start: which will be the first Fizz-Buzz number? Write your guess on your whiteboard.

Explore (about 8 minutes)

Pairs use 24 square tiles to build every different rectangle that uses all the tiles (a 3 by 8 and an 8 by 3 count as the same). They record each one as a multiplication and decide how they know they have found them all.

Predict: You have 24 square tiles. You must use all of them to make one rectangle. How many different rectangles do you think you can make? Write your guess.

- Stuck: start with one long row. How many tiles in the row? Now try 2 rows.
- Not sure if 4 by 6 and 6 by 4 are different: turn your rectangle round. Is it the same shape?
- Finished early: do it again with 36 tiles. Something special happens. What is it?

Mini-lesson (about 5 minutes)

Factor pairs in order: 36 = 1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6 (5 does not divide 36); stop when the pairs meet and draw a factor rainbow. Common factors: 12 apples and 18 oranges share the factors 1, 2, 3, 6, so 6 bags with 2 apples and 3 oranges each. Common multiples: buses every 6 and 8 minutes meet again after 24 minutes. Before page 2: square numbers have an odd number of factors (100 has 9; 18 has 6); for riddles, list the multiples of the bigger number first, then check each clue.

Worksheet answers (questions 1 to 4)

Q1	(a) 1 × 48, 2 × 24, 3 × 16, 4 × 12, 6 × 8 (b) 1, 2, 3, 4, 6, 8, 12, 16, 24, 48: 10 factors (c) 7 is not a factor, and the next number, 8, is already in a pair (6 × 8). The pairs have met, so there are no more.
Q2	6 ways: 2 rows of 20, 4 rows of 10, 5 rows of 8, 8 rows of 5, 10 rows of 4, 20 rows of 2.
Q3	(a) 18: 1, 2, 3, 6, 9, 18. 24: 1, 2, 3, 4, 6, 8, 12, 24 (b) 1, 2, 3, 6 (c) 6 packs, each with 3 red and 4 blue beads.
Q4	(a) Bus 33: 6:00, 6:10, 6:20, 6:30 … Bus 51: 6:00, 6:15, 6:30 … Together again at 6:30 am. (b) Every 30 minutes: 6:00, 6:30, 7:00, 7:30, 8:00, so 5 times.

Talk time

Ali says: "Bigger numbers always have more factors." Is he right? Give an example to show it.

Answer: Ali is wrong. 24 has 8 factors but 29, which is bigger, has only 2 (1 and 29). 36 has an odd number of factors because one of its pairs is 6 × 6, and 6 is only counted once.

Exit ticket (about 3 minutes)

Write all the factors of 28. Then find the smallest number that is a multiple of both 6 and 9.

Answer: 1, 2, 4, 7, 14, 28. The smallest common multiple of 6 and 9 is 18.

Watch for these mistakes

- Listing factors at random and missing one, or stopping too early (for 48, forgetting 6 × 8), instead of trying 1, 2, 3 … in order until the pairs meet.
- Mixing up factors and multiples: saying 12 is a factor of 4, or listing 4, 8, 12 as the factors of 4.
- For common multiples, multiplying the two numbers every time (10 × 15 = 150 minutes) instead of listing the multiples to find the first one they share (30).

Extension for fast finishers

Which numbers from 1 to 50 have exactly 4 factors? Is there a pattern? (There are 15: 6, 8, 10, 14, 15, 21, 22, 26, 27, 33, 34, 35, 38, 39, 46.)

After the taster

Tell the parent one thing their child noticed and one thing they explained well. Then share the next steps: [dates, fee and sign-up link]. The free diagnostic for the matching level helps choose between Level 4 (ages 9-11, covers P4 topics) and the level either side (see the placement note).